

CS595 – Medical Informatics & AI
Assignment 4 – Week 4 Reflections

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In the first four weeks of CS595 Medical Informatics and AI, I have come to understand the area of healthcare technology and how AI can be implemented in it. I have understood that AI in healthcare is closely linked to ethical accountability, which makes it one of the most crucial and high-stake areas of AI implementation since it has a lot of risks that are involved.

Before coming into computer science I wanted to pursue in the medical field, I see this as an opportunity to contribute. Among the most crucial topics that I have learned is the importance of Electronic Health Records (EHRs) and interoperability standards to current healthcare. I understood that healthcare data is not kept in a single clean dataset, but rather exists in a number of systems which have to communicate safely. It also made me realise how a simple mistake can make a lot of confusion and cause lawsuits. Learning about FHIR (Fast Healthcare Interoperability Resources) made me realize how healthcare organizations are transitioning to modern web based APIs that are constructed around resources like Patient, Observation.

A particularly interesting topic was SMART on FHIR, which enables third-party applications to integrate securely with EHR systems through standards like OAuth2. I found it meaningful that these technical mechanisms are not only software features, but also a foundation for compliance with privacy regulations such as HIPAA and GDPR. I like the fact how everything is made to work alongside each other to make the process more efficient.

The other major lesson has been the knowledge of how the privacy laws in healthcare are crucial to a patient. Prior to taking this course, I was familiar with HIPAA in a broad sense, but this course has given me the opportunity to recognize how privacy regulations limit the extent to which patient information can be used.

The other topic that I found interesting was the importance of Health Information Exchange (HIE) in facilitating the process of sharing data among various healthcare facilities in a safe manner. I also learnt that the patients usually have more than one provider, which includes hospitals, specialists and primary care clinics, and their records must be available in all these locations to maintain continuity of care.

It is clear from the class that AI used in healthcare is so much different from other industries and domains. A successful model that can be statistically performed well in one field can be deemed successful. In the medical field, a minor mistake may have a severe consequence, including wrong clinical judgment, bias, or invasion of privacy. The

course has highlighted that the concept of ethics determines the way AI serves humanity by applying principles such as patient safety, fairness, accountability without any bias.

Altogether, the first four weeks have changed my attitude towards AI in medicine. AI is no longer perceived as a replacement of clinicians, but rather, I view it as a decision support tool that should be implemented responsibly into the healthcare systems. The most influential AI apps will probably be aimed at enhancing the working process, assisting compliance, decreasing the administrative load, and improving patient safety without losing trust.

The casestudy and discussion based approach has proved to be quite useful in terms of the course structure. It has enabled me to relate technical ideas like interoperability, consent, and governance to the real life healthcare limitations. The exposure to more anonymized datasets or system-level implementation examples would be one of the areas where learning can be further enhanced.

I also learned that the extent to which healthcare AI relies on transparent and explainable ai systems is more than I had expected. Most AI tools can work well under controlled settings, but in practice in a clinical environment and doctors must have some understanding of why they are being given a recommendation before they can trust it. This puts in place issues of explainability, bias reduction, and a requirement of human control in decision making.

Another lesson that I learned during the course is that healthcare AI should continue to be patient-focused and not technology-focused. Contrary to other sectors where efficiency is most likely to be the main objective, in healthcare, a decision directly impacts human lives, and thus AI systems have to be created to promote improved outcomes, avoiding infringement on patient dignity and trust. I came to know that the terms informed consent, patient autonomy, and human control are important in the process of implementing AI in clinical settings.

I have found the class sessions so understandable and interesting, in particular, due to the numerous discussions that were backed with experience of industry professionals. Listening to real problems encountered in healthcare institutions made me relate the concepts to what was really happening in the hospitals. It is also due to this exposure that I am more interested in pursuing opportunities in the area of healthcare AI.

I am very interested and looking forward to learn more industry grade technologies and how they are being used in the remaining of the semester.